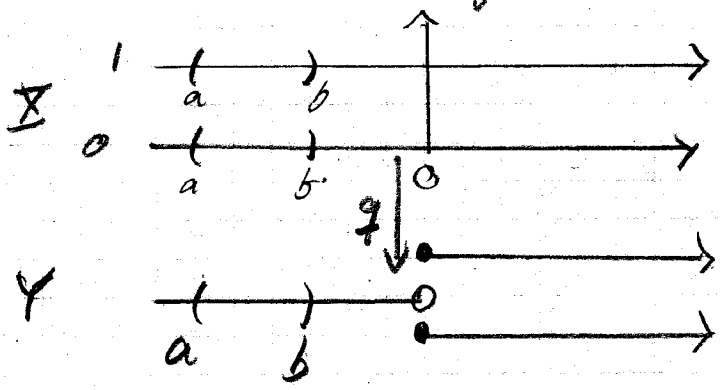
New spaces from old ones

Suppose $q: X \to Y$ is surjective. The map q is called a quotient map if it satisfies the condition: (Φ) $U \subset Y$ is open if and only if $q^{-1}(U) \subset X$ is open. In this situation, the topology on Y is called the quotient topology.

Note: The quotient topology is the strongest topology in which q is continuous.

Typically one uses condition (Φ) to define a topology, that is, given a surjetive $q: X \to Y$, where X is a topological space, define a topology on Y by (Φ).

Ex: Let $X = \{(x,0) \mid x \in \mathbb{R}\} \cup \{(x,1) \mid x \in \mathbb{R}\} \subset \mathbb{R}^2$ equipped with the subspace topology.
Form Y by identifying $(x,0) \sim (x,1)$, $x < 0$,



and let $q: X \to Y$ be the natural projection. Give Y the quotient topology.

Then $(a,b) \subset Y$, $a<b<0$, is open. A nbhd of $(0,1) \in Y$ is a union of the intervals $(c,0)$ and $\{(x,1) \mid 0 \le x < d\}$.

Note that $(0,0), (0,1) \in Y$ can not be separated by nbhds, so Y is not T2.

Recall: In general, $f^{-1}(Y \setminus B) = X \setminus f^{-1}(B)$.
Thus, when $q: X \to Y$ is a quotient map,
$C \subset Y$ is closed iff $q^{-1}(C) \subset X$ is closed.

Universal property: Let $q: X \to Y$ be a q.m., and let $f: X \to Z$ be any function.

$$\begin{tikzcd} X \arrow[d, "q"'] \arrow[dr, "f"] & \\ Y \arrow[r, "\tilde{f}"'] & Z \end{tikzcd}$$

Suppose f respects q, that is, if $q(x_1) = q(x_2)$, then $f(x_1) = f(x_2)$. Then there is $\tilde{f}: Y \to Z$ s/t $f = \tilde{f} \circ q$. Moreover, f is cont. iff $\tilde{f}$ is cont. (prove it!)

Let $X = \dot{\bigcup} A_\alpha$ be a partition of X. (Recall that partitions are the same as equivalence relations.)

Form a new space $X/\sim$ from X by "collapsing" each A_α to a point. Hence a point in $X/\sim$ corresponds to a set in X.

Ex: $X = [0,1]$, and partition X by $X = \{0,1\} \dot{\bigcup}_{0<x<1} \{x\}$ (so the endpoints are identified).
Then $X/\sim \; = \bigcup_{0<x<1} \{x\} \cup \{p\}$, where $p \sim \{0,1\}$.

As you can see, $X/\sim$ is a circle.

write $[A_\alpha] \in X/\sim$ for the point corresponding to the set $A_\alpha \subset X$.

Let $q: X \to X/\sim$, $q(x) = [A_\alpha]$, where $x \in A_\alpha$. Typically one equips $X/\sim$ with the quotient topology.

Ex: Let $X = \{(x,0) \mid x \in \mathbb{R}\} \cup \{(x,1) \mid x \in \mathbb{R}\} \subset \mathbb{R}^2$, and partition X by $X = \{0\} \cup \bigcup_{x \neq 0} \{(x,0),(x,1)\}$.

Then $X/\sim$ is the real line with two origins, which is a simple example of a non-Hausdorff space (show it!).

You can construct many other examples in the same vein.

Let $q: X \to Y$ be a q.m. Call a set $U \subset X$ saturated if $q^{-1}(q(U)) = U$.

If $Y = X/\sim$, then $U \subset X$ is saturated iff U is the union of some collection of A_α's.

Note that $O \subset Y$ is open iff $O = q(U)$, where $U \subset X$ is open and saturated. (If $O \subset Y$ is open, then $q^{-1}(O) \subset X$ is open and saturated. Now $O = q(q^{-1}(O))$ since q is onto.

On the other hand, if $O = q(U)$, where $U \subset X$ is open and saturated, then $q^{-1}(O) = q^{-1}(q(U)) = U$, so $O \subset Y$ is open. (Check all this!)

Prop. In the diagram where q is a q.m. $\tilde{f}$ is a homeomorphism iff

$$\begin{tikzcd} X \arrow[d, "q"'] \arrow[dr, "f"] & \\ Y \arrow[r, "\tilde{f}"'] & Z \end{tikzcd} \qquad \text{so } f = \tilde{f} \circ q$$

(i) f is cont. and surjetive

(ii) $f(x_1) = f(x_2) \Rightarrow q(x_1) = q(x_2)$ (injectivity of $\tilde{f}$)

(iii) $f(U) \subset Z$ is open for all saturated open $U \subset Z$ ($\tilde{f}$ is an open map)

Note that (iii) is not needed if Y is cmpt and Z is T2.

Ex: Let $X = I = [0,1]$ with $0 \sim 1$ only. We have

$$\begin{tikzcd} I \arrow[d, "q"'] \arrow[dr, "f(x) = e^{i2\pi x}"] & \\ I/\sim \arrow[r, "\tilde{f}"'] & S^1 \end{tikzcd}$$

Now $f(x)$ respects q so $\tilde{f}$ exist. Clearly f is surjective, and $f(x_1) = f(x_2) \Rightarrow x_2 = x_1$ or $x_2 = 1$, $x_1 = 0$ (or v.v.).

Now $I/\sim$ is cmpt as I is cmpt. It follows that $\tilde{f}$ is a homeomorphism.

Ex: Let $D \subset \mathbb{R}^2$ be the open unit disk and $\bar{D}$ its closure. Let $e^{i\varphi_1} \sim e^{i\varphi_2}$ for all φ_1, φ_2, so $\bar{D}/\sim$ denotes the closed unit disk with the boundary collapsed to a point. We will show that $\bar{D}/\sim$ is homeomorphic with S^2 (the unit sphere).

Let $b: D \to \mathbb{R}^2$ homeomorphically (i.e., $b(p) = \frac{p}{1-|p|}$).
Recall that the stereographic projection $\psi: S^2 \setminus \{N\} \to \mathbb{R}^2$ provides a homeomorphism between the punctured sphere and the plane.
Thus $c = \psi^{-1} \circ b$ yields a homeom between D and $S^2 \setminus \{N\}$.

Define $f: \bar{D} \to S^2$ by $f(p) = \begin{cases} c(p), & p \in D \\ N, & |p| = 1 \end{cases}$.

Claim: f is continuous.
For this, let $\mathcal{U} \subset S^2$ be open. If $N \notin \mathcal{U}$, then $f^{-1}(\mathcal{U}) = c^{-1}(\mathcal{U})$, which is open in D, and thus in $\bar{D}$. If $N \in \mathcal{U}$, we compute:
$\neg f^{-1}(\mathcal{U}) = f^{-1}(\neg\mathcal{U}) = c^{-1}(\neg\mathcal{U})$, which is cmpt in D, and also in $\bar{D}$. Hence $\neg f^{-1}(\mathcal{U}) \subset \bar{D}$ is closed, so $f^{-1}(\mathcal{U}) \subset \bar{D}$ is open also in this case.

Now in the diagram
$\tilde{f}$ obviously exist, and one argues as in the prev. ex. that it is a homeom.

$$\begin{tikzcd} \bar{D} \arrow[d] \arrow[dr, "f"] & \\ \bar{D}/\sim \arrow[r, "\tilde{f}"'] & S^2 \end{tikzcd}$$